Sachin Sivesh R K

B.E Electronics and Communication Engineering
+91 8778141841 | sachinsivesh23@gmail.com
LinkedIn | GitHub | LeetCode | **Languages:** English, Tamil, Kannada

Education

PSG Institute of Technology and Applied Research
2024 – 2028
Bachelor of Engineering – ECE
CGPA: 9.17/10 (upto 4th semester)

Vidyaa Vikas International School
2024 | 2022
Class XII: 94.4%
Class X: 93.0%

Technical Skills

- **Programming Languages:** Python, C, SQL, Verilog, MATLAB.
- **Technologies and Libraries:** Machine Learning, Deep Learning, Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, Plotly, Streamlit, Excel.
- **Areas of Interest:** Digital System Design, Circuit Analysis, Data Structures and Algorithms, Database Management Systems, Computer Networks, Operating Systems, IoT.
- **Soft Skills:** Leadership, Teamwork, Articulate Communication, Problem Solving, People Management.

Projects

On-Chip TDET/PDET Sensor Placement Optimization SanDisk Cerebrum 2026

- Developed a data-driven optimizer for placing thermal and process sensors on an SoC from timing paths, using percentile-based thresholds and enforced spacing limits.
- Modeled power-vs-speed trade-offs as separate adaptive voltage scaling operating points, keeping benefit estimates in the 8–14% range seen in industry field data.

Advanced Disruption Sensing Agents (SAP Tools) SAP Hackfest 2026

- Built a multi-agent system on SAP HANA Cloud and SAP Business Application Studio that monitors operational and external data signals to detect supply-chain disruptions early.
- Assessed disruption impact and recommended mitigation actions, enabling faster, data-driven decisions.

NeuroVision AI: MRI Brain Tumor Classification ↗

- Developed a deep learning-based system for MRI-based brain tumor classification using CNN architectures including ResNet and EfficientNet.
- Applied transfer learning and data augmentation to improve model generalization across glioma, meningioma, pituitary and no-tumor classes

Publications

Biometric Multimodal Authentication System Using Fingerprint and Finger Vein Recognition
Journal Paper – Under Review, Signal, Image and Video Processing (Springer)

- Developed a deep learning-based multimodal biometric authentication system integrating fingerprint and finger vein recognition using ResNet and ArcFace models.
- Enhanced authentication accuracy, security and spoof resistance through hybrid feature fusion.

Achievements

- **First Runner-Up** at **SAP Hackfest.** **September 2026**
- **Finalist** at the **SanDisk Cerebrum Innovation Bazaar.** **October 2026**
- **Top 30 Semi-Finalist** in the **Caterpillar Tech Challenge**. **May 2026**
- **Top 25 Finalist** in the **Qualcomm Edge AI Hackathon.** **June 2025**
- **Finalist** in the **iTech Ideathon** (Flagship Hackathon of PSG Institute of Technology and Applied Research). **May 2024**
- **Finalist** in **CODE-O-CLOCK** 24-Hour Hackathon. **September 2024**

Course Certifications

- **Verilog for an FPGA Engineer with Xilinx Vivado Design Suite** – Udemy
- **MATLAB Onramp** – MathWorks

Leadership & Activities

Vice President – iTech Intellect Quiz Club (Previously Joint Secretary, 2025 – 2026) 2026 – Present
Joint Secretary – ECE Association, Department of ECE (Previously Executive Member, 2025 – 2026) 2026 – Present
Volunteer – National Service Scheme (NSS), PSG Institute of Technology and Applied Research 2024 – Present
Media Head – FESTO (FATC), PSG Institute of Technology and Applied Research 2024 – 2025